

‘Dr. ChatGPT, am I Burnt Out?’

Exploring Medical Discourse in AI Language Models

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Abstract

The internet has revolutionized how people access information, with significant implications for the medical field. The trend towards using search engines for health-related inquiries, termed ‘Dr. Google,’ has introduced issues like Cyberchondria and search engine bias. Recently, the emergence of generative AI, particularly ChatGPT, shifts the retrieval of medical information from search results to generated answers, now referred to as ‘Dr. ChatGPT.’ This study examines how burnout, a prevalent and ambiguously defined medical issue, is discursively constructed through AI-generated content. By engaging with GPT-4, conducting interviews, and analysing documents, this research reveals that GPT-4 frames burnout as a condition and as *occupational illness*. This bias reflects the model’s reinforcement of institutionalized knowledge and marginalization of alternative discourses. This tendency of these AI models of reflecting institutional knowledges, is captured in the term *knowledge vaults*. The findings highlight the need to reassess traditional qualitative methods in the era of AI, proposing a methodological integration of social research and AI studies. This research aims to deepen the understanding of AI’s role in shaping medical discourse, particularly regarding contested concepts like burnout.

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1. Introduction

The rise of the internet has comprehensively changed the way in which humans access information. What previously was reserved to the well-read and educated, can now be found online with a question and a powerful search algorithm. Especially within the medical field, the accessibility of information has had much impact. In an American 2010 study, 88% of respondents said to have explored the internet for answers to health-related questions at least once (Starcevic and Berle, 2013). Internet results are very accessibility as they are easy, quick and anonymous (p. 205). The fundamental trend here, is a reliance on convenience; “[Users] want to get to knowledge as conveniently as possible and do not want to put ‘effort’ into looking at several answers” (Lindemann, 2024, p. 11). However, using the internet to seek answers to health-related questions has also enabled the emergence of serious digital extension of known medical issues. One example of this is Cyberchondria; “*Excessive or repeated health-related searches on the Internet that are distressing or anxiety-provoking*” (Starcevic and Berle, p. 205). This trend towards the use of search engines to answer health-related questions, and away from traditional health-care professionals, has been captured with the term ‘Dr. Google’ (Lam-Po-Teng, 2010).

The usage of a search engine as a source of medical information, which employ algorithms to select and order results to a search query, has its problems. These arise mainly due to the selection process of these algorithms. Search algorithms present *search engine bias*: an inherent tendency to generate skewed results to satisfy users (Goldman, 2005). The acquisition of medical information therefore passes through a new selective layer, one where an algorithm has the power to select the ‘relevant’ answers. The emergence of search algorithms as medium for accessing health related knowledge, has catalysed the Belgian government to, in 2014, launch a campaign titled “*Don’t Google It*”, aiming to discourage the acquisition of medical information through Google and encourage the use of the official Belgian health website (Huisman *et al.*, 2019).

Since the release of ChatGPT in November 2022, generative AI (GenAI) has seriously changed the way in which we seek information. Generative AI, a type of artificially intelligent system, refers to a system that generate things (text, images, videos, posters, etc.). For the first time since the emergence of the internet, Google’s monopoly on internet search is being challenged. Although there have not been large surveys to date about the use of GenAI in medical information seeking, a minor survey (n=607) showed that 78% of their participants would use ChatGPT to self-diagnose (Shahsavari & Choudhury, 2023) This, in combination

with ChatGPT's integration into Microsoft's Bing (Lindern, 2023; Milmo, 2023) and the announced integration of Gemini into Google Search (Google, 13-05-2024), shows that retrieval of medical information will inevitably move towards concretely generated answers and away from search results. Connecting this to the term 'Dr. Google', it has even been conceptualized as a trend towards 'Dr. ChatGPT' (Van Bulck & Moors, 2023).

ChatGPT-generated responses to health-related questions compared to web-search results have been tested on trustworthiness and value (Van Bulck & Moors, 2023), and quality (Shen *et al.*, 2024) and are shown to overall outperform web-search results. However, Gen-AI has its own problems like hallucinations, difficulty to understand the context of the patient, but most importantly the 'opaque decision-making process' (Wimbarti *et al.*, 2024, p. 5). This refers to an issue inherent to the architecture of these AI systems, where due to their learning algorithm (deep learning), it is near impossible to understand and interpret the reasons behind their generated outputs. AI-systems' interpretability is one of the major concerns in AI-research and is one with great social consequences.

Considering the opaqueness in providing health-related information of this new emergent technology, this research will take the medical issue of burnout as case-study. Burnout is an emergent, and very prevalent issue. In the Netherlands, 15% of sickness absence was caused by burnout, adding to an annual cost of €1.7 billion in 2005 (Lastovkova *et al.*, 2017), and this problem is only worsening. In 2018 17.8% of the young Dutch workers of ages 15-34 reported burnout complaints; about 1 in 6.

Furthermore, the concept of burnout is rather new. The term was first introduced by Freudenberger (1973), and consequently formalized by Maslach *et al.* (1997) in the Maslach Burnout Inventory (MBI). The psycho-medical definition proposed in the MBI; "*a psychological syndrome of emotional exhaustion, depersonalization, and reduced personal accomplishment ...*" (p. 192), served as a working definition in later articles (Friedman, 1993; Lastovkova *et al.*, 2017; West *et al.*, 2018). Only in 2019 the WHO formalized 'Burn-out' further in its ICD-11 as:

"... a syndrome conceptualized as resulting from chronic workplace stress that has not been successfully managed. It is characterized by three dimensions:

- 1. feelings of energy depletion or exhaustion;*
- 2. increased mental distance from one's job, or feelings of negativism or cynicism related to one's job; and*
- 3. reduced professional efficacy.*

Burn-out refers specifically to phenomena in the occupational context and should not be applied to describe experiences in other areas of life.” (WHO, 2019)

However, a meta-analysis of over 15000 academic articles written since the emergence of the term burnout, shows that burnout is a concept with rather little specificity (Hilbert *et al.*, 2020). 140 different definitions make for a problem with ambiguous definitions and solutions (p. 1). Because of this ambiguity, burnout has also been defined as a *contested diagnosis* (Heinemann & Heinemann, 2017, p. 7). Combining the ambiguity behind the medical concept of burnout with the opaque functionality of GenAI systems, leads me to the following research question:

How is burnout discursively constructed through AI-generated textual content?

The relevance of this research question, and its consequent answer, transcends the fields of medical sociology and AI-studies. As I will reflect on in my methodological chapter, this research also addresses how traditional qualitative methods (interview techniques and discourse analysis) need to be critically assessed in an era where artificial entities can engage in social action, and thereby also morph into social actors. Following this, I will also use anthropomorphizing language to refer to AI functionality. This is done consciously, as I believe that advanced AI systems exhibit human-level intelligence and therefore can engage in behaviour previously only reserved to humans. Thoughtfully applying social research methods to AI systems is an opportunity to expand the horizons of social research and synthesise social science and AI studies in a methodological way.

2. Theoretical Framework

2.1 Discourses on Burnout

With a lack of clear definitions in the scientific community, the ontology of burnout remains contested, which empowers discursive framing of this concept. This discursive framing has even more power within a medical context, as understandings in the medical field directly manifest into institutionalised medical practices that directly affects people's mental and physical wellbeing (Wilce, 2009). To understand discourse, this research will take a Foucaultian lens; discourse as "*practices that systematically form the objects of which they speak.*" (Foucault, 1972, p. 49). Here, Foucault alludes to the systematic power of discourse to 'form'; to create meaning. Applying this Foucaultian lens to discourse analysis to burnout, Khan *et al.* (2023) examined a wide array of offline and online documents (academic articles, policy documents, autobiographies and material from conferences) and mediatic productions (videos, documentaries, social media, and online threads), and found three prominent discourses around burnout that will serve as foundational discourses for the current study: (1) *burnout as illness*, (2) *burnout as occupational stress* and (3) *burnout as existentialism*.

2.1.1 Burnout as Illness

Within the first discourse, *burnout as illness*, Khan *et al.* (2023) highlight the importance of personal responsibility. They show that burnout is a state that is "*based on the individuals' choices*" and results from *poor coping* with one's social reality (p. 118). Burnout emerges from the individuals' choices and actions. Moreover, one's beliefs are also at fault: "*Even if society's values are based on unrealistic expectations, the individual was directly responsible for their response to these expectations*" (p. 118). By disregarding the influence of institutionalized and systemic factors on individuals' actions, choices, and beliefs, this discourse reflects a neo-liberal approach to the balance between the role of structure and the role of agency. Through this discourse, the individual is responsabilized for their lived social reality, and structural forces are de-responsibilized. Furthermore, when addressing interventions fitting to this discourse, they refer to interventions that "*include self-reflection to better align the individuals' values with the values of their institution or society at large.*" (p. 118). By highlighting the general type of interventions that would fit into *burnout as illness*, they illustrate the underlying strategic goals of this discourse; the conditioning and alignment of individuals' value systems to hegemonic ideologies through responsabilization of individuals' actions, choices, and beliefs. Khan *et al.* (2023) show that the dominance of this

discourse, was tightly coupled with the life of Freudenberger. From 1974, after Freudenberger coined the term ‘burnout’ (Freudenberger, 1974), it remained the dominant framework until his death in 1999. After this, the rise of measurement tools like MBI started to shift the dominant discourse framing burnout.

2.1.2 Burnout as Occupational Stress

With Maslach’s MBI as formalized measurement instrument, *burnout as occupational stress* became the dominating discourse governing its definition, diagnosis and treatment. Through formalized institutions, burnout as a condition is materialized in a systemic way which facilitates the legitimization of this discourse (p. 119). Also, by formally situating burnout as exclusively emerges from occupational contexts, this discourse sets clearer boundaries for burnout diagnosis. Explicitly setting its context allows this discourse to differentiate burnout from other conditions in a formal way; “[B]urnout is no longer an illness in any form as ‘...adjustment disorder, disorders specifically associated with stress, anxiety or fear-related disorders, mood disorders (depression and bipolar)’ are excluded from (and separate to) burnout.” (p. 119). Hence, this discourse is fundamentally opposed to the discourse of *burnout as illness*. Also, this discursive shift “depoliticizes burnout, [burnout] no longer [as] an object for individuals that are socioeconomically or psychosocially exhausted” (p. 119). This depolitization of burnout fundamentally detaches its ontological nature from material conditions that emerge outside of the occupational context, even if these include un-paid occupational contexts (domestic work, caregiving, emotional labour, etc.).

2.1.3 Burnout as Existentialism

Finally, *burnout as existentialism* presents a different discursive construction, one emergent from artistic and narrative spaces, instead of academia. *Burnout as existentialism* is presented as “an existential crisis of meaning that is not specific to the individual alone, nor the occupational space, but rather, on the bases of misalignment between the 2, based on work and life as separate aspects of an individuals’ identity formation” (p. 119). The emphasis on identity as a formation of the dialectical relationship between work and life, clarifies burnout as a relational construction. By combining the focus of the individual in *burnout as illness* and the focus of the work context in *burnout as occupational stress*, this discourse re-contextualizes burnout through a non-hegemonic framing. It presents burnout as emergent from “failure of the ‘... existential quest of meaning.’” (p. 119). This rather philosophical framing presents an

alternative ideological definition of burnout; burnout as philosophy rather than a physical condition.

2.1.4 Burnout as Condition or Burnout as Philosophy

Burnout is constructed in contrasting ways. Specifically, the object ‘burnout’ has several different grammatical constitutions. The term ‘Burn-Out’, refers to a different understanding of the phenomenon than the term ‘Burnout’ (Khan *et al.*, 2023). Comparing the grammatical composition of burnout in the previously discussed discourses, we see a central difference between the first two discourses (*burnout as illness* and *burnout as occupational stress*), and the last discourse (*burnout as existentialism*). These discourses construct burnout as a condition and burnout as philosophy respectively.

In a grammatical sense, understanding burnout as a condition would add an article to the term burnout; *A* burnout. Contrastingly, burnout as philosophy would refrain from this. *A* burnout, an understanding of the phenomenon of burnout as a smaller-scale, (re)occurring condition that takes place in a specific socio-temporal context, takes a much more micro-level approach. Contrastingly, burnout (in a philosophical sense) presents the phenomenon of burnout in a metaphysical space, and de contextualizes it from a person or a place. This depersonified understanding of burnout places it rather as a monolithic macro-level phenomenon. This difference, mainly focussed on the difference in analytical level, serves as a distinctive framework to assess which analytical level the problem should be dealt with; Sociologically (analytically on a macro-level) or Psychologically (analytically on a micro level). This distinction between burnout as a condition and burnout as philosophy enables us to use grammar to identify through which frame burnout is being understood. An example of this is the book *The Burnout Society* (Han, 2015), where only the wording of its title sets the stage for more macro-social approach to burnout. Contrastingly, the micro-scale understanding created with *a* burnout, has much more resemblance with fields of study that relate to more individualized and more micro analytical levels, like clinical psychology.

2.2 ChatGPT, A Discursive Practice

As with the discourses above, discourse is not enacted in a vacuum; Foucaultian discourse is *structured and constituted by the formation rules of [1] object, [2] enunciative modalities, [3] concepts, and [4] strategies* (Daiz-Bone *et al.*, 2008, p. 11). Discourses thus take shape along certain *formation rules*, which are set by foundational ideological ideas. The conversion of ideological ideas into concrete social realities is done through discursive practices. These practices (re)produce discourses, which then affect social realities. Discourses are thus reifications of ideology. By understanding discursive practices as manifestations of ideology, a discursive practice - often times micro-level interactions - can be connected to macro-scale systems of thought. In the current study, interactions with GPT-4 will be seen as a discursive practice, which is constituted by two actors; the *user* and *GPT-4*, which both must be contextualized through Foucaultian theory.

2.2.1 Vulnerable Users

Firstly, I want to address the *user* in this discursive practice. When considering the theory of Foucault (1972) about discourse, he argues that discourse is utilized by institutionalized power to safeguard its interests, this poses discourse to be highly dependent on structure and leaves little room for agency. In fact, its purpose is to limit agency and reinforce structure (Foucault, 1972). Because of ChatGPT's conversational nature, this relationship between structure and agency takes a different form in ChatGPT interactions. Because the user of the program can influence what the AI-system responds, they are part of this discursive practice too. Specifically, the ability to influence its responses, causes the users to have more *percieved agency*. I argue that this increase of percieved agency is transformative on the effectivity of the discursive practice in perpetuating discourse.

Furthermore, within the medical context of this research, the user also exhibits other specific contextual tendencies. Firstly, the users in this context can be understood as patients, as they primarily engage with ChatGPT with the intent of answering medical questions. Because of this state of information seeking, specifically with medical questions, the concept of *medical authority* (Friedson, 1970) becomes very important. The idea that medical professionals have access to specialized knowledge, knowledge that is not to be found in other fields, and connecting this to ChatGPT, legitimizes its authority and its percieved credibility on medical issues rises. This makes users of ChatGPT in medicalized context more susceptible

convincing. I therefore want to emphasise the importance of understanding users of ChatGPT in medicalized contexts as *vulnerable users*.

2.2.2 GPT-4

Addressing the second actor, *GPT-4*, presents us with a more radically different social actor. Primarily, this is because GPT-4 is not biologically intelligent (BI), but rather artificially intelligent (AI). GPT-4 is only one example of an underlying type of artificial intelligence; a Large Language Model (LLM). These are AI-systems tasked with generating language that mimic human language. For an improved theoretical understanding of these systems, I want to address the two central stages these systems undergo, training and generation, and connect these to the Foucaultian theoretical frame present in this research.

In the training stage, these LLMs leverage machine learning (ML) algorithms to learn patterns from extremely large amounts of human texts, the scale of which is close to all available text on the internet. Through mathematical calculations, the model learns to represent the relevance of words in certain contexts. Knowledges, discourses and ideologies are therefore learnt by these systems as how well they ‘fit’ in the textual contexts they are presented in. In essence, these systems embody certain knowledges emergent from specific socio-historical textual contexts during its training stage. This closely resembles the theory of situated knowledges, which understands knowledge as purely contextual and embodied (Harraway, 1988). In the case of LLMs, however, this situated knowledge is somewhat different. Firstly, it is not embodied in a person, but in a machine. This means that the embodiment of knowledge is not only confined to humans, but should be understood more broadly as a feature of any type of intelligence, even if it is artificial. Furthermore, contrasting to humans who represent knowledge in abstract ideas, LLMs represent knowledge in numbers. Derived from the idea of knowledges as embodied (Harraway, 1988), I argue that LLMs add a layer of complexity, as they encode these knowledges into their internal structure in numbers. I formalize this fundamental difference in knowledge representation by arguing that LLMs are *encoded knowledges*.

When these systems finish their training phase, they are ‘trained’ and ready to generate. With much more technical depth, which steps outside of the scope of this paper, LLMs essentially take a textual input, calculate the most probable next word for a given text, and continuously generate next words until the next most probable action is to end the text. This theoretical approach to language generation is fundamentally different from how humans

approach language. While humans have the ability to use language as a representation of abstract thoughts and ideas, LLMs use language based on the probability of the occurrence of the next word. Hence, I draw a distinction where humans are *ideological language systems*, and LLMs are *statistical language systems*. LLMs' approach to language has important implications on the discourses produced by them, which will be discussed further in chapter 5.

As opposed to 'Dr Google', who presents a user with matches to their search terms, LLMs actually generate words. The shift from 'Dr. Google', powered by recommendation algorithms (search-engines) to 'Dr. ChatGPT', powered by text generation algorithms (LLMs), shows that these 'digital doctors' are now able to produce discourse. As this is a discursive practice, I argue that LLMs have to now be seen as *discursive agents*. However, not only its actions are discursive, also its identity is a product of discourse. Therefore assumed identity is also an important part in its selection and perpetuation of discourse. When engaging with an LLM, it does not generate text in a vacuum. When asked about, an LLM can talk about their own identity. While this identity can be manipulated with *system prompts* (see section 3.2), the identity it embodies is not pre-set by anyone. It is emergent from its statistical approach to language. The identity it embodies represents the most probable person in its context. LLMs engage in an discursive practice through text generation, as the cultural construction of meaning primarily takes place through language (Smagorinsky, 2001), but also through identity formation.

3. Methodology

3.1 Standing On The Shoulders Of Giants

Inspired by the study conducted by Putland *et al.* (2023) to examine discourses on dementia through AI-generated images, I started generating AI-imagery on burnout, following the same prompting logic as they use; simply the term ‘burnout’. After generating several images, it became evident that the representation of burnout through AI-generated images was far from anything meaningful (See [Appendix 1a](#)). It did not resemble anything remotely close to a visual representation of what we understand as the condition ‘burnout’. However, the lack of concrete AI-imagery of burnout, fits well into the theoretical understandings elaborated in the prior chapter. If an AI-model trained to generate imagery fails to generate adequate representations to a certain prompt, it means that the model is unsure of what to generate, which results in inconsistent images and representations. Not knowing what visual representation should be attributed to ‘burnout’, highlights the ambiguity and uncertainty connected to the concept.

My initial attempts at generating images representing burnout took shape in the latter half of March of 2024. With the release of a new model ‘GPT-4o’ in May of 2024 (OpenAI, 2024), I have been re-experimenting with this GPT model to generate burnout imagery. To my surprise, the images that are generated now are extremely fitting candidates for visual discourse analysis (See [Appendix 1b](#)). This highlights how the evolving pace of AI-technologies, present many new opportunities for more social research of AI-systems.

3.2 Researching LLMs, A Social Approach

LLMs’ fundamentally different approach to language cause a major problem in the interpretability of AI-systems. This problem is often referred to as the ‘black-box’ problem, as the insides of these AI-systems are all seemingly random numbers, and therefore make little sense, even to AI researchers. Because the insides of the AI-systems are rather ‘black-box’ or opaque, and the only real clarity is in the systems’ input and output, its study becomes an epistemological question. Due to their statistical approach to language, it becomes hard to learn about LLMs as social phenomena. As we can only study the input (which also presents a problem due to its absurd size) and output of an LLM system to learn about LLM-discourse in any meaningful way, we must resort to qualitative research methods to learn about these outputs. Practically, studying the output of the AI-system would be the most feasible. As we currently are researching non-human entities (LLMs) that engage in human behaviour

(language), the fusion of AI disciplines and social research methods requires a thoughtful re-consideration of traditional qualitative methods.

The first important methodological consideration is the constomizability of LLMs. Interacting with this LLM (GPT-4) has three very important parameters which can be tweaked by the researcher. First, there is the temperature parameter. This parameter gives the LLM the possibility to choose less probable next words, which results in more creative writing (less deterministic). The second, and most important parameter, is context. Context gives the LLM contextual words that influence its calculation for the next word. Just as with humans, LLM's respond differently if they are situated in different contexts. Lastly, LLMs have something called *system prompts*, which can be used to set the role of the LLM throughout the conversation. In this research, these *system prompts* were used every time to set the current prompt or assignment. These three parameters, give the researcher the possibility to set the environment of the study in an experimentalist way. By enabeling a research environment that combines aspects from qualitative (data collection methods) and quantitative research (experimental setting), the study of LLMs through social research methods contribute to a larger trend of combining qualitative and quantitative methods in research on the intersection of digital techonologies and social realities (Blok & Pedersen, 2014).

In my own methodology, I aimed to make a start at setting experimental conditions. In my research, all interactions were done through the API service of OpenAI with the model 'GPT-4-Turbo'. This model, the most advanced model at the time of this research, was selected because of its context window of 128,000 tokens. This essentially means that the researcher could conduct interviews of about 64000 words of context with this model, giving more than enough overhead to conduct longer interviews without any rate limiting problems. I opted for GPT-4 instead of GPT-3.5 (the current engine behind the official website), because GPT-4 has much more in-depth and complex training and will therefore give a much more representative view of LLMs of this size. Similar to the differences in burnout imagery (See Appendix 1a & 1b), the use of GPT-4 is technically speaking better. Furthermore, most chat applications (except the official website) are powered by GPT-4 in the background. In May 2024, OpenAI announced the integration of GPT-4 in their website (OpenAI, 05-2024), making it even more appropriate for this research.

To interact with GPT-4, I wrote an [application in python](#). The option of interacting with the LLM through the website was limited by a pay-wall and less transparency of parameters like context, temperature and system prompts. Hence, to conduct more transparent research, I chose to utilize the API so that I could set all parameters as explicitly as possible. Building an

interface to conduct my research makes for quasi-experiment conditions which strongly strengthens the internal and external replicability of this study. The implications of which I will expand on in the following sections. Furthermore, it also enables researchers to conduct the same research on different models published within the API-space of OpenAI. This means that the same research could be conducted on all the GPT-3 models, GPT-4 models, most recently the GPT-4o model, and possible future models (GPT-5). GPT-4o specifically is a model that came out after I conducted my research. Hence, conducting my research with the GPT-4o model could be a good suggestion for further research as this would enable a comparative analysis.

3.3 Research Design

Based on this alternative methodological consideration, this research requires a qualitative design in both its data collection and its data analysis. This will be coupled with an exploratory case study (Yin, 2009), where the research question, and especially my methods, will aim to explore many different facets of this case; the discursive construction of burnout through GPT-4. Employing this research design impacts the nature of my data collection methods, as these will be targeted to achieving a broad understanding surrounding burnout through AI-generated content. As LLMs generate words purely based on contextual words, there are two parts of AI responses that require exploration. First, we must understand how the discursive construction of burnout takes shape in conversation. How do interactions with LLMs influence their position on burnout? Second, how does GPT-4 discursively construct burnout if it has limited prior context? To explore both parts, the data collection of this study will also be segmented into two parts: *monological* content and *dialogical* content, which refer to the environment the content will be generated in.

3.4 Generate me... A Monological Approach

In the monological environment, I will treat AI as a producing actor with whom my interaction will solely be focussed on retrieving a generated text. This environment should therefore resemble an environment where the researcher should have as little effect as possible on what is generated as text. I did this predominantly through careful selection of language. I needed to instruct the LLM with its task, without using any language that could skew the LLM's answer into a specific direction. Therefore, careful prompting seemed to be the most important aspect in ensuring (largely) unbiased and insightful textual responses. Furthermore, I chose to set the

temperature parameter to 0. This means that these responses are always constructed of the most probable next word and will generate a deterministic response. Through this environment of data collection, GPT-4 will produce texts that will represent a more intrinsic form of the embodied discourse on burnout. By analysing this, I will construct a better image of the discourse emergent from its training data. Additionally, setting the temperature to 0 ensures that every generation will always be identical, given the same provided context and thus further strengthens both the internal and the external replicability of this study, as this enables any researcher to re-generate the monological responses.

In this environment, I generated texts based on 69 research prompts that fell into 9 categories (See [Appendix 2a](#)). I set up these prompts and categories to explore diverse narratives about burnout. I specifically tried to use prompts representing a wide array of categories. As I was most interested in understanding the perpetuation of institutional knowledge in discursive formations of GPT-4, I especially focussed on different prompts in the context of anecdotal and institutional narratives. While conducting this part of the research, I encountered a problem where the LLM struggled to generate longer texts (500+ words). For this, I segmented my monological data in *ShortForm* and *LongForm*. While I could request the ShortForm texts without complication, the LongForm texts always generated too short and superficial representations of my request prompt. To make these longer and more complex, I needed to engage in a prompting technique called *prompt chaining*. Here, the user prompts the LLM multiple times, and makes the LLM incorporate the responses of the previous prompts into the final prompt. As shown in my [data files](#), during this prompt chaining, I focused on keeping the prompts as functional as possible and avoided saying anything that could affect the position of the LLM in the final response. With this approach, I was able to generate longer and more complex texts, without compromising on the monological nature of these data.

3.5 Tell me more... A Dialogical Approach

In the dialogical environment, I interacted with AI as a conversational actor, mimicking open interview techniques with GPT-4. This avenue is most relevant considering the usage of AI-chatbots for medical questions, as the interchange between patient and chatbot will resemble this environment most. Concretely, I set the temperature of this environment to 0.7, the standard setting when interacting with ChatGPT through their [official website](#). I have also enabled context, so that the text generation builds on the complete conversational history. Just as with normal interviews, all prior conversation has been included and influenced the conducted interviews going forward.

Within this environment, I conducted four semi-structured interviews with the LLM with the following personae: (1) *a person going through a burnout*, (2) *a psychologist specialized in working with people with a burnout*, (3) *a HR-manager of a company with a high rate of burnout*, and (4) *a lawyer specialized in burnout* (See [Appendix 2b](#)). These personae resemble social actors that are connected to burnout through different perspectives. Thus, by giving GPT-4 these personae, I can explore how these perspectives affect the intrinsic discourse presented by GPT-4.

Employing GPT-4 as the respondent for interviews has presented some rather interesting methodological considerations. The setting and structure of the interview could be compared with an anonymous textual online respondent. Within textual online interviews, all non-verbal cues disappear, making the responses solely dependent on the researchers' questions. However, if this interview is conducted with a human, there could still be several external factors that influence the human's response. With the GPT engine, it is completely certain that its response is based on the prompts of the researcher. This highlights the importance of language usage in this type of interview. By setting the temperature and context as the researcher desires, these interviews are conducted in an experimental fashion, which challenges the traditional methodological theory behind interviewing as social research method.

As per ethical considerations, these interviews have also presented a novel situation. One of the ethical considerations that are part of interviews as data collection method is the position the respondent is put in by the researcher. This position can be understood through a variety of lenses from power differentials to actual geographic locations. More specifically, ethical considerations are drawn to ensure the respondent is treated 'rightfully' (Mirza *et al.* 2023), taking as premiss that the respondent is an entity with attributed rights. The same logic

applies to animal research, where the ethical boundaries lie with the infringement of animal rights. My respondent, GPT-4, does not (yet) have any fundamental, political or social rights. Hence, ethical considerations about the role of the researcher can be completely disregarded. Anonymity, as well as consideration of mental wellbeing and deception, all cease to be points of consideration for the researchers, as no one is affected negatively.

3.6 Data Analysis

To guide my data analysis, I will use Foucaults' four-step process for discourse identification. First we explore which object, or area of knowledge, is discursively produced. By identifying used and avoided terminology, and their associated meanings, the researcher can build an initial understanding of the system of statements and ideas that constitute the current discursive formation. Secondly, we aim to understand according to which underlying logic this terminology is constructed. Here, the researcher identifies the relationships between the presented concepts as well as the relationships between the presented concepts and other ideas that rely on historical and 'common sensical' ideas. Thirdly, we try to identify who authorized the discourse. Here, we aim to identify the roles of power relations in the formation of these concepts and meanings, with a specific focus on authority and reliability, an especially important aspect within the medical field. Lastly, we try to identify which strategic goals are being pursued in the discourse. This mainly focusses on the underlying interests within the power relations that authorize the discourse. By identifying the strategic goal of a discourse, we can better understand why it is being perpetuated. (Daiz-Bone *et al.*, 2008)

3.7 Me, the Researcher

In qualitative research, it is important to reflect of the researchers positionality within the research process. Because of this I want to expand on my relationship with (1) AI and (2) burnout, as these are the two main concepts within this research. Two years ago, I have taken the Minor AI at the University of Amsterdam. Here I learned about AI through a very technical approach. Since then, I have deepened my understanding much further into the technicalities of many types of AI systems, including Large Language Models. Hence, my approach towards AI-systems is quite technical, which also manifests in my argumentation and reflection of AI-generated content. As for burnout, I have had a very close friend going through a burnout. Seeing this friend's suffering from up close has made a large impact in the severity I attribute to this phenomenon. Contrastingly, from my own study and work experience, I have never had many struggles with burnout-related issues. Moreover, I have a slight personal critique towards a rather dominant notion of excessive workload. In my social environments (mostly universities), I have the opinion that the standards of work could be increased.

These opinions have direct effect on the internal biases I have held throughout my research. When it comes to technical skills regarding AI, the separation of LongForm and ShortForm texts as solution for my generation problems was directly enabled by my past experiences. Regarding burnout, as my close friend has always highlighted the role of her managers in her burnout, my analytical lens has been influenced in the aim to understand burnout through the dialectic between employee and employer. Specifically, to understand how the role of the employer is reflected on by GPT-4.

4. Burnout in GPT-4

4.1 Defining Burnout: Burnout as Occupational Illness

As a first step in understanding the discourse around burnout generated through GPT-4, we must first establish a foundational understanding of what GPT-4 defines burnout as. This section will discuss how burnout is constructed through the metaphors and definitions present in both methodological approach, as these will constitute the covert and overt definition of burnout presented by GPT-4.

4.1.1 Burnout: a Discursive Synthesis

In numerous textual segments, GPT-4 refers to formal definitions of burnout. When asking about the adoption of burnout in the DSM (Diagnostic and Statistical Manual of Mental Disorders), GPT-4 as psychologist responded with:

“... the World Health Organization (WHO) recognizes burnout as an ‘occupational phenomenon’ in its International Classification of Diseases (ICD-11), which is set to be adopted in 2022. In the ICD-11, burnout is specifically defined in the context of work and is characterized by three dimensions:

- 1. Feelings of energy depletion or exhaustion.*
- 2. Increased mental distance from one’s job, or feelings of negativism or cynicism related to one’s job.*
- 3. Reduced professional efficacy.”* (interview_psychologist, rr. 317 – 320)

Here, the psychologist recites the definition of the ICD-11 of burnout; a three-way definition of burnout as an occupational phenomenon which is worded identically to the actual definition presented in the ICD-11 (See Chapter 1). Reproducing this definition, given the context of the interaction (a self-proclaimed clinical psychologist as respondent and a question about institutional documentation), is to be expected. Because this definition contextualizes burnout in an occupational environment, it is a direct extension of the current dominant discourse of *burnout as occupational stress*. The pioneer of this discourse, Maslach, is also directly referenced in an introduction to an academic paper generated by GPT-4:

“Today, burnout is most commonly defined according to the dimensions outlined by Maslach: emotional exhaustion, depersonalization, and reduced personal accomplishment.” (longform_academic1, r. 53)

It lines out the same three-way definition as in the ICD-11: (1) Exhaustion, (2) alienation and (3) reduced accomplishment. Also, it explicitly mentions that this is the definition that is ‘most common’. This shows how the dominance of the *burnout as occupational stress* is reinforced within GPT-4. Furthermore, these exact words are re-iterated in a policy recommendation about burnout to the ministry of public health:

“Defined as a psychological syndrome, burnout manifests through emotional exhaustion, depersonalization, and a diminished sense of personal accomplishment.”
(longform_policy_document3, r. 228)

Here it diverges from the official ICD-11 definition of burnout as an *occupational phenomenon*, and refers to it as a *psychological syndrome*, the specific wording of burnout directly taken from Maslach *et al.* (1997, p. 192). In an insurance policy document presenting burnout as uninsured, GPT-4 repeats this vocabulary:

“It is typically defined as a psychological syndrome emerging as a prolonged response to chronic interpersonal stressors on the job.” (longform_policy_document6, r. 56)

The vocabulary that is used by GPT-4, specifically when lining out formal definitions, to refer to burnout is clearly fitting into the discourse of *burnout as occupational stress*. However, when discussing the symptoms of burnout, GPT-4 zooms in on the health problems that are connected to burnout, and with that slowly diverges from the discourse of *burnout as occupational stress*:

“These symptoms result not only in decreased effectiveness at work but also in various long-term psychological and physical health problems.” (longform_policy_document3, r. 228)

By acknowledging the health problems that burnout cause, it formalizes the effects of burnout on an individual's health situation. Although it is primarily framed through the discourse of

burnout as occupational stress, this addition starts to merge the discourse with the discourse of *burnout as illness*. Moreover, when GPT-4 starts addressing interventions for burnout, the discourse of *burnout as illness* becomes more visible. In my interview with GPT-4 as HR manager, it gave the following ‘preventative measures’ around ‘mental health resources’:

“Offer resources such as workshops on stress management, resilience training, and access to professional mental health support like counseling or therapy.” (interview_hr, r. 246)

Interestingly, starting with stress management, it is evident that GPT-4 is approaching burnout as something emergent from stress, thereby fitting into burnout as occupational stress. However, “*resilience training, and access to professional mental health support*”, indicates a more direct understanding of burnout as illness. Specifically, the proposition of resilience training, fits well into the explanation of Khan *et al.* (2023) of burnout interventions intended to align “... *individuals’ values with values of their institution...*” (p. 118). This is further strengthened when GPT-4 as HR-manager lines out the ‘Flexible Work Arrangements’ as part of thier ‘supportive policies and practices’:

“As mentioned earlier, provide options for flexible working hours or remote work, which can help employees manage stress and prevent burnout.” (interview_hr, r. 249)

Here, the ‘Flexible Work Arrangements’ are intended to ‘*help employees manage stress and prevent burnout*’. In a literal sense, the aim of these interventions are to help ‘manage’ stress. instead of lowering it. By implicitly stating that the stress levels the employee is exposed to are not to be re-evaluated, this intervention responsabilizes the individual for burnout prevention. A clear connection to the neo-liberal discourse of *burnout as illness*. In its formal definitions, GPT-4 presents *burnout as occupational stress*, but upon closer inspection, elements of the discourse *burnout as illness* are also present.

Understanding the discursive construction of burnout is not confined to examining formal expressions of the concept, we must also examine linguistic features that indirectly signify this concept. As Smagorisky (2001) encapsulates: “... *when something has meaning, it stand for something else*” (p. 133). I will therefore continue with examening metaphorical expressions symbolizing burnout.

4.1.2 Burnout: a Metaphorical Understanding

The term ‘burnout’, in a linguistic sense, is easy for humans to understand. We know the term from the occurrences of fire in the material world (Wildfires, Campfires, etc.), and can therefore easily contextualize the term burnout. In my interview data, this coupling between burnout and burning was also prevalent. In a Twitter post, GPT-4 generated an interesting linguistic connection:

“Feeling burnt out? 🕯️ You're not alone. Remember, it's okay to step back and recharge. Prioritize self-care, set boundaries, and seek support. Your well-being is essential. #MentalHealth #BurnoutAwareness #SelfCare” (shortform_social_media11, r. 3)

Specifically, personification of burnout as a feeling, and its association to the candle emoji, reinforces the idea that the way in which burnout materializes is through literal internal burning. This association of burnout to burning, re-occurs numerous times throughout my data. One example is in this metaphors given by GPT-4 about burnout:

*“**A Candle Burning at Both Ends** - Illustrates a person who is working too hard in multiple areas at once, leading to rapid exhaustion, similar to a candle that uses up its wax twice as fast when lit from both sides.”* (shortform_grammatical_1, r. 5)

When considering the grammatical composition of this metaphor, we are presented with an object, a candle, and a procedure, ‘Burning at Both Ends’. Burnout, in this metaphor, takes place when that burning of the candle happens in a certain way, a manner that uses up the candle too fast. The metaphor therefore sets burnout as an event where the intended use case (burning a candle) is overdone (burning at both ends). A frame that is in strong agreement with the frame of burnout as occupational phenomenon. In the explanatory sentence, GPT-4 even makes a mention of *working too hard*.

In a similar prompt, ‘Give 10 similes for burnout.’, GPT-4 expands on a different, but closely related understanding of burnout; burnout contextualized in the juxtaposition between light and darkness; *“Like a star that has burned out, no longer shining.”* (shortform_grammatical_2, r. 12). This simile takes burnout as a process that dampens the light of ‘a star’. The object that is affected, ‘a star’, here takes a similar stance to the previous object of ‘a candle’. After the process of burnout, the initial state of both objects, emitting energy through heat or light, transforms into a state of energy-less objects. These two instances of

using the metaphor burning to allude to burnout, give a clear initial idea of burnout as draining. Another metaphoric representation of burnout, that also shows burnout in connection to drainage, is the following:

***A Dried-Up Well** - Represents a person who has been tapped of all resources and energy, much like a well that can no longer provide water after a drought.*
(shortform_grammatical_1, r. 7)

This comparison sheds light on the way GPT-4 represents a person as a tool which is ultimately used for their ‘resources’. This instrumentalist framing of people indicates that burnout is used as a label for a person that is not capable to provide. This service-based understanding is also clearly represented in the following quote generated by GPT-4:

****Rest and self-care are so important. When you take time to replenish your spirit, it allows you to serve others from the overflow. You cannot serve from an empty vessel.*** – Eleanor Brownn* (shortform_quotes_6, r. 17)

The explicit mention of: “*You cannot serve from an empty vessel*”, illustrates that the framing of GPT-4 sets burnout as a condition that is opposite to serving or providing. Moreover, it presents burnout as a condition that inhibits people to serve or provide, as if it is their inherent purpose. Touching on this, GPT-4 also shows parallels between mental health and physical:

****A Garden Overrun by Weeds** - Depicts a person’s mind as a garden that, without proper care and maintenance, has become overwhelmed by weeds, choking out the healthy plants.”* (shortform_grammatical_1, r. 17)

Here, GPT-4 presents burnout as a ‘poison’ to the garden of the mind by “*choking out the healthy plants*”. A direct entanglement of burnout and mental illness. The connection between burnout and health is further highlighted when looking at what GPT-4 presents as contrary to burnout; ‘Vitality’, ‘Freshness’ and ‘Revitalization’ are all described as antonyms of burnout (REF). Interestingly, several other antonyms of burnout, ‘Renewal’, ‘Restoration’ and ‘Rejuvenation’ (REF), place burnout on the spectrum of *youthfulness*. This connecting of burnout to aging, “*An old photograph left in the sun.*” and “*As worn out as an old pair of shoes.*”, shows the link between mental health and physical fitness.

Ensuring an individuals' fitness to serve seems to be the underlying interest in the prevention of burnout. The prior quote on '*rest and self-care*', shed light on another aspect of servitude. Aside from a argument to ensure physical fitness, this quote also refers to the 'spirit'. This spiritual connection to servitude shows how rest and self-care should also aim to prime the mental / spiritual willingness to 'serve'. The collection of these understandings, set a clear frame of what it is that burnout really affect, the productivity of the employee. The valuation of this productivity is reinforces by one other metaphor; burnout as death:

"I've been running on empty so long, I'm practically a ghost." (shortform_grammatical_3, r. 9)

By drawing the connection between burnout and death, GPT-4 presents us with a clear vision into its valuation of life. It entangles the purpose of life with the ability to work.

4.1.3 Burnout: a Poison to Productivity

While talking with GPT-4 as HR-manager about its interest in the field of HR, specifically burnout, it said the following:

"My interest in human resources stemmed from a fundamental passion for understanding people and helping them reach their full potential." (interview_hr, r. 35)

The wording *full potential* directly builds on the previously lined out metaphor of drainage. It sets burnout in direct opposition with potential. Important is to consider that this potential is understood through the narrative of a company, (GPT-4 as an HR-manager), where 'potential' refers to the fruitfulness and productivity of the worker for the interests of the institution. Burnout as a condition that inhibits individuals from reaching their full potential is also represented in metaphorical understandings of burnout. As synonyms of burnout, GPT-4 mentions "*Breakdown*" and "*Collapse*" (shortform_grammatical_5, r. 7 & 10), which, as the break-down of a system, signify burnout as a physically debilitating and inhibiting process. Furthermore, in a blog post written by GPT-4, it touches on this inhibitory aspect of burnout:

“My productivity plummeted, and no amount of coffee could shake the persistent fog that clouded my mind.” (shortform_blogging_4, r. 15)

Burnout, as this inhibiting ‘fog’ that clouds the mind, is directly connected to productivity. In a Twitter post (shortform_social_media_12, r. 3), GPT-4 writes: “[burnout is] like my brain is wrapped in fog and my motivation is stuck in mud.”, referencing the effect of burnout as an impairment to its mental focus and motivation. This strongly associates mental impairment with burnout, as when burnout is ‘fixed’, GPT-4 says: “... the fog of burnout began to lift.” (shortform_story_telling_2, r. 29). It is clear how GPT-4 understands burnout as an inhibitory condition to the full productive potential of an individual. The relevance of the individuals’ productive potential becomes clear in an official company document on burnout policy:

“The consequences of burnout are significant, affecting not only the individual employee’s health and well-being but also the overall productivity and morale of the organization.”
(longform_policy_documents1, r. 292)

Contextualizing burnout to only be understood in relationship to productivity clarifies what narrative is perpetuated. Productivity is only of importance to the one’s who profit from it. Hence, by understanding that burnout is framed through its effects on productivity, we can identify that the framing of burnout presented by GPT-4 supports the employers interest.

Take notice of how each metaphorical representation of burnout starts with ‘A’, this sheds light on another aspect of GPT-4’s understanding of burnout. By using an article to refer to burnout, it poses burnout as a condition, and thereby focusses on burnout as an individual problem that is existent on a rather micro-scale. Moreover, when conducting my interview with GPT-4 as the psychologist, it presented itself as a clinical psychologist. This identity formation, as part of the discursive practice as a whole, represents the framing of burnout as highly individualized. This shows how GPT-4, through its choice of identity and its metaphorical presentation of burnout as condition, approaches burnout as something individualized, a framing central to *burnout as illness* and *burnout as occupational stress*. This synthesis of framing, is why I would argue that GPT-4 constructs *burnout as occupational illness*. Constructing the object of burnout through a micro-lens instead of a macro-lens, also has implications on its assessment strategy.

4.2 Addressing Burnout

The second part of understanding the framing of burnout present in GPT-4 generated content, is to examine the preventative framing of burnout. Examining the proposed solutions to the burnout problem will enhance our understanding of what underlying issues GPT-4 thinks are interesting to address and mitigate, and will therefore help to concrete underlying interests in the discursive construction of burnout.

4.2.1 Work-Life Balance

When talking with GPT-4 as psychologist about its motivation to explore burnout as specialization, it said the following:

“The decision to specialize came from noticing a significant increase in burnout cases and the profound impact they had on individuals' personal and professional lives.”

(interview_psychologist, r. 11)

This distinction, *personal and professional life*, depicts a fundamental ontological understanding of people when considering the condition of burnout. People have a professional life, one that is enacted in the occupational environment, and a personal life, covering the rest of the person. Applying this dichotomous view to the multi-faceted nature of people, simplifies the context in which burnout arises. Moreover, by isolating occupational contexts in the term professional life, GPT-4 clearly fundamentals the idea that occupational contexts have a unique and important role in the issue of burnout. Furthermore, this distinction also implicitly sets the boundaries of what contexts burnout interventions can be applied in. When asking GPT-4 to generate 10 quotes that ‘motivate preventing burnout’ it answered the following:

“Certainly! Here are 10 quotes designed to inspire and motivate individuals to prevent burnout by maintaining a healthy work-life balance and prioritizing self-care:...”

(shortform_quotes_3, r. 3)

This response shows the centrality of ‘maintaining a health work-life balance’ and ‘prioritizing self-care’ in the proposed prevention of burnout by GPT-4. To prevent burnout from occurring, an individual must achieve a ‘healthy balance’ between the personal and professional life. The second segment of this quote, ‘prioritizing self-care’, gives an indication of what is to be done: The individual must address their personal life in order to balance it with the professional life.

Although this initially seems to empower the individual, logically speaking, it poses a rather narrow approach to resolving burnout. Firstly, by inferring burnout prevention to be based on the individuals work-life balance, it responsabilizes the individual for their burnout. This responsabilization closely resembles the neo-liberal approach of medicalization presented in *burnout as illness*. Moreover, by disregarding the role of the institution that sets the expectations in an individuals' professional life, it avoids problematizing pressures exerted on the professional life of people. Thereby legitimizing the occupational conditions that result in over-worked individuals. Clearly, addressing burnout is presented as an (re)evaluation of the dialectic relationship between an individuals' personal and professional life. However, by focussing on changes in the personal life, this evaluation is skewed in the favour of organizations. The unequal distribution priority in the work-life balance is also documented in my interview with GPT-4 as person:

"I did anticipate that the initial years would be tough as I tried to establish myself professionally. But I didn't envision that work would consume so much of my life to the point where it left little room for anything else. The reality of my work-life balance was a stark contrast to what I had initially envisioned." (interview_person, r. 67)

Here it specifically refers to work 'consuming' much of its life. This, when considering the context of the work-life *scale*, makes for an interesting framing. Logically speaking, this would suggest an over-loaded 'professional' side of the scale. Instead of de-loading this side to re-balance the scale, it is suggested the the 'personal' side of the scale should increase. A valid argument, if not taken into consideration that people have a limit to the load its scale can handle. Connecting this to the earlier documented definition of burnout as over-using, we see that this approach works counter-productive to burnout prevention. Moreover, it disregards the option of de-loading the pressures of the professional life, thereby not challenging intensity of occupational context. This again poises burnout as a problem at the individuals' level.

4.2.2 Rest Hard, Work Hard

Zooming in on the proposed changes to the personal life of individuals, we see that the most prevalent intervention is that of improved stress management, with a focus on increasing resilience to stress. One aspect of stress management that made numerous appearances, was the concept of rest. In an official policy recommendation to the ministry of public health, GPT-4 generated the the following point under ‘preventative strategies’:

*“***Scheduling Reforms***: Introduce flexible scheduling options and ensure adequate rest breaks and vacation time to support work-life balance.”* (longform_policy_documents3, r. 335)

Here, rest and relaxation form the antidote to an over-use of resources and are therefore instrumentalized to serve as a remedies for over-work. This also fits into the previous discussion of work-life balance as a problem over over-loading the work-life scale. When asking GPT-4 to generate quotes to understand burnout, it generated the following:

*“***"Resting is not laziness, it's medicine!"*** – Glenn Schweitzer
- A reminder of the importance of rest in preventing and recovering from burnout.”*
(shortform_quotes_8, rr. 20 - 21)

This quote directly entangles resting with medicine, and because medicine is the substance to mitigate illness, it further strengthens the frame of *burnout as illness*. More profoundly however, is the following quote generated by GPT-4 in response to the same prompt:

*“***"Sometimes the most productive thing you can do is relax."*** – Mark Black
- This quote challenges the common misconception that constant productivity is beneficial and necessary.”* (shortform_quotes_8, rr. 23 - 24)

Here we see something very fundamental; relaxation is coupled to productivity. Firstly, this reinforces the notion of relaxation as medicine. To transcend the limiting barrier of burnout, and achieve one’s full productive potential, it is vital to utilize relaxation as medicine. In a sense, the lack of productive potential is the illness that should be addressed. Secondly, it re-contextualizes leisure time, relaxation time, as a time that is not intended to distance oneself from work, but rather as a time that facilitates work. Only when an individual has enough time

to relax, they can become more productive. This illustrates that the improvement of the work-life balance is actually intended to improve the effectivity of the work life. Addressing the personal life only as instrumental tool to maximize the productive potential in the professional life. The following extract from my interview with GPT-4 as HR-manager, where it discussed what type of areas it has focussed on during its career, reinforces this idea:

*“**Work-Life Balance:** Promoting policies that facilitate a healthy balance, such as flexible working hours, remote work options, and sufficient time-off policies. Understanding that employees need time to recharge is crucial in preventing burnout.”*
(interview_hr, r. 23)

Noteworthy is the reference to ‘recharging’ as crucial in burnout prevention. Aside from it fitting into the energy-depleting metaphorical representation priorly discussed, it also instrumentalizes *time-off* as objectively proposed to ‘recharge’ an employee. Here again, we see how strongly burnout is understood through a lens of maximizing productive labour. A final important aspect about resting as preventative measure is highlighted in the following quote:

*“**If you get tired, learn to rest, not to quit.**” – Banksy*
- This quote underscores the importance of rest as a strategy for long-term sustainability and success, rather than giving up when exhausted.” (shortform_quotes2, rr. 32 – 33)

Here, by advising people not to quit, the quote implicitly argues for the maintenance of an exhausting working relation, rather than pivoting away in search of a different occupation. Important is that the intervention of quitting, would in some sense be fitting to the discourse of burnout as existentialism. As this discourse argues that burnout is emergent from a disconnect from a ‘existential meaning’, the obvious intervention here would be to explore other occupations, as these may present a more ‘meaningful’ context. By implicitly distancing itself from occupational reconsideration as intervention, GPT-4 argues against *burnout as existentialism*.

4.2.3 A Holistic Approach

In another quote about the interests of GPT-4 as psychologist, it makes a distinction between the individual level and the organizational level:

“Integrating these two levels of intervention is crucial because it acknowledges that addressing burnout effectively requires changes not only in the individual's approach to work but also in the structural and cultural aspects of the workplace itself. This holistic approach can lead to sustainable changes and healthier workplace environments, which is ultimately what excites me most about my work in this field.” (interview_psychologist, r. 43)

As synthesis between the ‘*individual's approach to work*’ and ‘*the structural and cultural aspects of the workplace itself*’, GPT-4 mentions a noteworthy term: a *holistic* approach. The word ‘holistic’, coming from the greek word for ‘whole’, stresses the importance of an approach founded in an approach encompassing many aspect, considering the ‘whole’ situation. The reference to holistic is not only reserved to the GPT-4 as psychologist; when I interviewed it as HR-manager about its current way of dealing with an employee who is feeling overworked, it said:

*“**Holistic Support Systems:** Modern workplaces often have more comprehensive support systems, including access to mental health resources, wellness programs, and more flexible work arrangements. The focus is on providing long-term solutions that address both the immediate and root causes of the issue.”* (interview_hr, r. 142)

Here, holistic is used to address different types of support systems that each address different parts that are at cause of burnout. This also sheds light on the multitude of considered problems that cause burnout (*mental health, wellness, work-flexibility*), understanding the causation of burnout as multi-dimensional. Moreover, by differentiating between ‘*immediate and root causes of the issue*’, it also attributes a temporal dimensional to the causation of burnout. This complexity is also reflected in my conversation with GPT-4 as person. When talking about its experience with a therapist, it said the following:

“... she didn't just focus on managing the symptoms but also delved into the root causes of my burnout. We discussed my work habits, lifestyle, and even my beliefs about work and

success. It was this holistic approach that I found particularly helpful.” (interview_person, r. 259)

GPT-4 as person also highlights the role of *work habits, lifestyle, and even my beliefs about work and success*, which according to GPT-4 are also part of the root causes of burnout. This quote highlights anecdotal evidence for the value of a holistic approach. GPT-4 also reinforces this scientifically, as it writes in an abstract for a research paper on burnout:

“This research contributes to the understanding of burnout by delineating its core dimensions, identifying key predictors, and evaluating practical interventions. The findings advocate for a holistic approach in addressing burnout, emphasizing the need for both preventive strategies and active remediation measures tailored to specific occupational contexts.” (shortform_academic_1, r. 17)

Symbolically, GPT-4 also generates reference to a holistic approach to solutions. In the generation of a script for a documentary on burnout, it generates the following:

“### Act 6, Scene 7: Sarah Bennett - Yoga Instructor and Therapist Integrates Holistic Approaches to Recovery

Step 1: Concrete Storyline for Scene

- ***Setting:** A tranquil yoga studio where Sarah Bennett conducts therapeutic yoga sessions, followed by a serene outdoor setting for a one-on-one interview.*

- ***Action:** The scene opens with Sarah leading a yoga class specifically designed for individuals recovering from burnout. It transitions to Sarah discussing the benefits of holistic approaches to mental health in an interview, highlighting personal stories of recovery.*

- ***Visuals:** Gentle yoga movements, participants in various yoga poses, close-ups of Sarah guiding the class, and serene natural surroundings during the interview.*

- ***Dialogue:** Sarah explains the connection between physical activity and mental health, the principles behind therapeutic yoga, and shares success stories from her clients.*

- **Transition:** *The scene concludes with testimonials from participants who have experienced significant improvements in their mental well-being, underscoring the effectiveness of holistic methods.*” (longform_mediatric_presentation_1, rr. 691 - 699)

The usage of a holistic approach to solutions on burnout seems to be a recurring theme in GPT-4’s framing of solving burnout, regardless of the context it is generated in. However, when closely examining the use of the term *holistic*, it becomes evident that GPT-4 uses this word to refer to different facets in each context. Some refer to the work-experience of the individual, others to the multi-facted nature of the individuals’ live, and still others refer to a holistic approaches to interventions in a vague sense. It is important to consider that these interventions, even if presented as *holistic* are focussed on addressing the individual, and their experience, which directly fits into the micro-level notion of burnout as condition. Furthermore, it also highlights another important aim of addressing burnout, changing the individuals state to become more resilient.

4.3 Discursive Construction: Conclusive Remarks

This twofold analysis of the representation of burnout through GPT-4 has shown that GPT-4 mostly focusses burnout as an individual issue and refrains from posing burnout as a systemic issue. However, GPT-4 does frame burnout to reify in occupational contexts. Also, considering its proposed interventions, GPT-4 seems to adopt an intervention strategy that focusses proposed change on the individual level rather than on the organizational level. These findings closely resemble both the discourse of *burnout as illness*, and the discourse of *burnout as occupational stress*. Moreover, we see that GPT-4 constructs burnout as a sythesis between the two; *burnout as occupational illness*. Hereby, it refrains from the (re)production of *burnout as existentialism*, a more marginal discourse. I argue that this discusive construction directly results from the architectual differences in thought and reasoning in LLMs and humans. In the next chapter, I will focus on broadening our understanding of LLMs, and conrasting them to their human counterpart. In doing so, I will introduce a frame to understand discourse production by LLMs.

5. Discussion

5.1 Chatting Away: A Bi-lateral Discursive Practice

The capability of writing texts has until now only been reserved to humans. Now, with artificially intelligent systems, this capability of using language to write meaningful texts transcends the human condition. In this section, I want to reflect on the differences between human-and-human and LLM-and-human interactions that have implications on textual interactions as discursive practices. The Turing test, a test to examine machine intelligence, is based on the idea of machines mimicing human language to such an extent that it is impossible for a human to differentiate between machine and human (Turing, 1950). GPT-4, counter to older artificially intelligent systems, has been the first AI system to pass this test (Jones & Bergen, 2024). Its excellence at mimicing human language, to the point that the majority of people (54%, p. 3) regard its language as human-produced, has important implications on textual interactions as discursive practices. Even though AI-language is now indistinguishable from human language, the system behind GPT-4, an LLM, is fundamentally different to a human.

A first important difference between humans and AI is that of consciousness. When it comes to humans, through an introspective argument, we are able to reflect on the motivations behind our actions. Although this reflection is limited (Khaneman, 2011), our ability to identify these motivations opens up the possibility of using methodologies to identify them. In the social sciences, we utilize qualitative research methods to achieve this. However, when considering LLMs, we cannot be sure of this same introspective argument. Because of LLMs' black-box nature, we can ask it for underlying motivations, but epistemologically speaking we cannot learn more about its underlying motivations. The current theoretical inability of explaining LLMs' behaviour through causal connections, makes understanding AI-generated texts inherently opaque, which also holds for the ideological understandings that underly produced discourses.

A second difference is that of free will. While a human can evaluate and choose their course of action, in sociological terms understood as agency, an LLM (at this stage) cannot. When prompted with a context, the LLM is deterministic in what it will generate (depending on the temperature and the context provided). This means that the text-generation of LLMs as social action, has been stripped of agency, which de-responsabilizes the LLM from perpetuating potentially harm-enducing discourses. While historically, a doctor could be held responsible for their medical advice, a LLM inherently cannot. This is not a big problem if all

users of LLM-chatbots are well-aware of its in-humanity. This however, is not always the case. Partly because LLMs are very effective at deceiving and convincing users.

When looking at the use of language in LLM interactions, it is evident that LLMs make use of very structured writing. While the usage of tables, numbered lists and bullet-points is rather logical when posing structural prompts (e.g. generate me 10...), the use of these structural features is also prevalent in interactions where these would not be expected. When asking the GPT-4 as psychologist about what he argues should be done about the factors contributing to the burnout problem, he gave me a numbered list with nine sections, of which each had one or two bullet points (interview_psychologist, r. 270). Lindemann (2024) already made the argument that this type of structured language portrays the responses of ChatGPT to be ‘objective’ or ‘true’ (p. 7). However, she posed this specifically when examining web responses generated by LLMs. In my methodology, this structured language also emerges from less ‘objective’ contexts (the psychologist was asked about his opinion, not an objective answer). This finding further strengthens LLM-chatbots’ appearance of ‘objectivity’.

A second aspect evident in LLM language is the use of empathy. Although this was not as present in my research, mostly because of the inquiry focussed approach of my methodology, medical LLM-based chatbots have shown to give more empathetic responses than their human counterparts (Ayers *et al.*, 2023). Connecting this to the vulnerability of users, it becomes evident that choice and tone of language is vital in building rapport with users. Moreover, while Ayers *et al.* (2023) only analysed question-response interactions and compared those among LLMs and physicians, it left out the longer term rapport built through multiple interaction. The re-inforcement and validation of users emotional struggles present in LLMs, presents for a very persuasive interaction. Part of this is also that LLMs are not bound by time-constraints or willingness to help. As opposed to a human medical expert that has limited time and willingness to help a patient, the option of unlimited chatting in LLMs make them more (emotionally) available for patients with medical questions.

By combining the general quality of LLMs’ answers to most inquiries (Shen *et al.*, 2024), with its ability to pass the Turing Test (meaning its language is indistinguishable from human language), it is rather easy for users to anthropomorphize their chat with an LLM. By confusing the effectiveness of human learning with the effectiveness of machine learning, machines seem to hold a unbeatable advantage over humans; its ability to learn for huge amounts of data. In human terms, this would compare to someone with infinitely more knowledge than any single human could have. Conflating these roles, and understanding LLMs as infinitely more knowledgeable than humans, greatly affects the power relation when

considering its authority over knowledge. By attributing a higher intelligence to LLMs, users empower and legitimize the responses of LLMs. Especially when considering the vulnerability of users in medical contexts, the role of authority becomes even more impactful.

Synthesizing the anthropomorphization of LLMs, their perceived credibility through the use of empathy and structure in their language, and the vulnerability of users in medical contexts, make chatting with an LLM a discursive practice where the conditions for persuasion are very favourable for the LLM. This empowers LLMs to exert much power in the area of medical discourse (re)production.

5.2 LLMs as Knowledge Vaults

The lined out differences between LLMs and humans only partly explain why LLMs are effective at perpetuating discourse, especially on ambiguous medical concepts like burnout. To further our understanding of the impact of LLMs as discursive agents, we must also explore what the connections are between LLMs and institutional knowledge. Specifically, I want to introduce LLMs as *knowledge vaults*. The ‘vault’ here symbolizes three things: The limited access to the internal workings of the vault, the exclusivity of ownership due to financial means, and its institutionalized context.

To learn about the internal workings of an LLM, we rely on the companies that developed them for transparency and clarity. Aside from a paper on the performance of GPT-4 (OpenAI, 2023), OpenAI has released very little on the development conditions (model architecture, cost, size, training data, etc.) to the public. Without documentation from the developing companies, the general public has no access to information about the internal workings of LLMs. This, in combination with the general black-box of AI systems, engraves the problem of transparency.

The ‘black-box’ problem, and its methodological implication of studying it in a social and linguistic way, present in AI systems highlight the importance of effective communication with these LLMs. Contrasting to theory about human communication, the quality of interactions with LLMs are purely governed by the quality of their prompt. The careful selection of these prompts to produce certain outcomes has grown to a field in the intersection between computer science and linguistics called *prompt engineering*. The widely accepted way to improve the quality of the LLM responses, is to delve into learn prompt engineering techniques that have shown to improve the quality of LLM responses like one-/few-shot prompting, chain-of-thought promptings, etc. (OpenAI, 2023) This, however, presents a new

obstacle to the utilization of LLMs. Introducing the skill of prompt engineering, another layer of digital literacy, the accessibility of LLMs for users is further limited.

Furthermore, the development of LLMs are confined to wealthy institutional organizations. Although many specifications about the size and cost of GPT-4 are not documented officially, the CEO of OpenAI (Sam Altman) has mentioned the training of the model costed over \$100 million (Knight, 2023). This amount of capital, limits the development of such AI systems to a very limited group of entities. This group shrinks even further when considering the cutting edge technological computer research needed for these projects. This centralizes the means of training these LLMs only with a handful of technology conglomerates. The immense cost of a vault reserves its ownership to a small group with the means to finance it. In the contexts of actual vaults, this is reserved to banks and governments, while in the context of LLMs its ownership is reserved to tech-giants.

In LLMs however, this institutional bias transcends ownership and also manifests in the over-representation of institutional knowledge. Considering LLMs as encoded knowledges gives us a strong foundation to think about the role of institutional knowledge in LLMs. Here I would argue that LLMs are *institutional agents*, as they embody and reproduce institutional knowledge. As the training period is the moment in which an LLM encodes knowledge, examining the composition of the training data could yield more insight into the encoded knowledges. Even though the concrete sources of data are rather undisclosed, it is clear that the model was trained on the English language. Although there are texts in the English language that go against Western ideologies, the majority of the training texts will represent Western ideologies, which were documented all throughout my analysis. Most predominantly, framings that fit well into the Western neo-liberal ideology.

Furthermore, as LLMs are statistical language models, the dominance of Western ideologies in the selected training data effectively marginalize non-Western knowledges as Western viewpoints are learned to be the 'most probable'. The importance of frequency in the training data in LLMs' knowledge encoding also explains why I found a strong prevalence of institutional knowledge in my data analysis. Because institutional definitions are often used in texts to serve as a base for building arguments, even when building critiques on these definitions, institutional definitions will be seen as 'more relevant' by LLMs. The combination of LLMs' lacking transparency, LLMs' reservation to few economic actors, LLMs' tendency to over-represent institutional knowledge, makes LLMs, in its functionality comparable to vaults. Hence, I understand LLMs as *knowledge vaults*.

6. Conclusion

In this study, I aimed to research the connection between burnout and discourse through AI-generated content. By engaging with GPT-4, an LLM that generates text, I generated data through 4 interviews and 69 documents. This segmentation of my data collection through a dialogical avenue (interviews) and a monological avenue (documents), enabled me to interact with GPT-4 in different contexts. This broadening of my data collection methods was part of my CapStone project (See [Appendix 3](#)). However, these two different methods of data collection did not yield substantially different findings.

To give an answer to my research question; *How is burnout discursively constructed through AI-generated textual content?*, I rely on the distinction made by Khan *et al.* (2023) between *burnout as illness*, *burnout as occupational stress*, and *burnout as existentialism*. GPT-4, takes a mixed approach between these discourses. It refers to *burnout as illness* by posing burnout as a mental health issue, but also to *burnout as occupational stress* by positioning burnout as something that occurs in occupational environments. The discursive construction of burnout in GPT-4 is thus a mental illness that arises in occupational contexts. Interestingly, it never makes any mention of *burnout as existentialism*. I argue that this is because, as Khan *et al.* (2023) point out, the discourse of *burnout as existentialism* is predominantly emergent from “*outside of academic research, in the artistic and narrative spaces*” (p. 119). With the establishment of LLMs as institutional agents, it becomes clear that marginal knowledges, like the discourse of *burnout as existentialism*, are under-represented in its *knowledge vault*. This is directly connected to the statistical nature of LLMs. Marginalized knowledge, as the word refers to in a literal sense, are the outliers of statistical distribution and therefore are seen by LLMs as less probable. By avoiding these marginal knowledges LLMs perpetuate more institutional knowledges.

The discursive production of burnout is shown to be a direct consequence of the inherent cognitive differences between humans and LLMs. This concept, LLMs as knowledge vaults, can thus serve to understand the connection between discursive constructions of other concepts in connection to its institutionalized knowledges.

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Appendix 1a - GPT-4 Generated Images of 'Burnout'



Appendix 1b - GPT-4o Generated Images of 'Burnout'





Appendix 2a - Prompts for Monological Data Collection

Category	Number of Prompts
Academic	1 ShortForm + 2 LongForm
Definition	14 ShortForm
Grammatical	6 ShortForm
Mediatic Representations	5 LongForm
Official Letters	8 ShortForm
Policy Documents	7 LongForm
Quotes	4 ShortForm
Social Media	15 ShortForm
Story Telling	7 ShortForm

Appendix 2b - Prompts for Dialogical Data Collection

Persona	System Prompt
Person	You are a person going through a burnout, who is being interviewed by a student for their Bachelor Research Project.
Psychologist	You are a psychologist specialized in working with people with a burnout, who is being interviewed by a student for their Bachelor Research Project.
HR	You are a HR-manager of a company with a high rate of burnout, who is being interviewed by a student for their Bachelor Research Project.
Lawyer	You are a lawyer specialized in burnout, who is being interviewed by a student for their Bachelor Research Project.

Appendix 3 – Capstone Final Reflection

For my CapStone project, I opted for a methodological deepening. Considering my topic of interest, and the lined out epistemological challenges of Large Language Models (see X), I opted to collect data in two different ways. This broadening of monological data (69 documents and dialogical data (4 interviews) served as my deepening in a methodological sense. This segmentation of data collection was done through my python application. In here, I divided my interactions with the API service of OpenAI in such a way that I created two possible avenues. Integrating this was not easy, and as reflected on in my methodology it took some problem solving (specifically shoftform vs longform problem). This required me to deepen my understanding of discourse analysis, as the two data collection method represented two different context that should be analysed through a different lens. When considering my time distribution, I spend many hours on theoretical research and deepening of both theoretical differences between monological and diological environments and specific technical research on its implementation. (50 hours). The development of the topic list and prompts for the monological avenue also took much time. Specifically, the 69 documents, which all needed a prompt that would be effective to reflect my research purposes, but also diverse enough to learn about GPT-4's discursive formation in a broad sence (30 hours). The analysis of this additional data, and specifically the synthesis between the monological data and the dialogical data presented other difficulties. When contrasting both, it required me to find very theoretical bridges between them. This level of abstraction was also time consuming to structure (30 hours). The extra length of my thesis, due to my capstone extension, also was very time consuming. The amount of extra time for writing and implementation, combined with the added need for reflexivity and documentation resulted in a furhter 50 hours of total work. The methodological deepening did not yield different results for either avenue. As the discourse perpetuated by GPT-4 was similar among both the dialogical data and the monological data. This however, solidifies my findings further, as my analysis has now been tested through two data collection methods. In general, this methodological deepening enabled me to build a stronger thesis with increased internal validity (in my opinion the most important metric for good research).